

PADHAISHURU

GATE Preparation Platform

GATE CS

Computer Science & Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 21 questions (32%)

Moderate: 36 questions (55%)

Difficult: 8 questions (12%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE CS — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

3M [-0.66]

[General Aptitude]

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is representative. Library 1st Main Road Canteen Physics Lab Hospital 5th Cross Road Chemistry Hostels Classrooms Lab N W E S

- (A) The library is located to the northwest of the canteen.
- (B) The hospital is located to the east of the chemistry lab.
- (C) The chemistry lab is to the southeast of physics lab.
- (D) The classrooms and canteen are next to each other.

Q2.

2M [-0.66]

[General Aptitude]

If ‘`’ denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered _____ daunted]. Which one of the given options is appropriate to fill the blank?

- (A) phased
- (B) phrased
- (C) fazed
- (D) fused

Q3.

3M [-0.66]

[General Aptitude]

In the sequence 6, 9, 14, x, 30, 41, a possible value of x is

- (A) 25
- (B) 21
- (C) 18
- (D) 20

Q4.

2M [-0.66]

[General Aptitude]

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}$ th of the number of students who like other branches. The number of students who like both their core and other branches is 500. The number of students who like their core branches is

- (A) 1,800
- (B) 3,500
- (C) 1,600
- (D) 1,500

Q5.

2M [-0.66]

[General Aptitude]

Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work. How many attempts does each wizard make before coming to this conclusion, independently?

- (A) 24
- (B) 48
- (C) 16
- (D) 12

Q6.

2M [-0.66]

[General Aptitude]

Consider the relationships among P, Q, R, S, and T: P is the brother of Q. S is the daughter of Q. T is the sister of S. R is the mother of Q. The following statements are made based on the relationships given above. (1) R is the grandmother of S. (2) P is the uncle of S and T. (3) R has only one son. (4) Q has only one daughter. Which one of the following options is correct?

- (A) Both (1) and (2) are true.
- (B) Both (1) and (3) are true.
- (C) Only (3) is true.
- (D) Only (4) is true.

Q7.

2M [-0.66]

[General Aptitude]

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube? Note: The figures shown are representative.

- (A) A
- (B) B
- (C) C
- (D) D

Q8.

2M [no -ve]

[General Aptitude]

Antonym of 'ephemeral':

Enter your answer here

Q9.

2M [no -ve]

[Engineering Mathematics]

Probability of at least one head in 3 coin tosses:

- (A) $1/8$
- (B) $3/8$
- (C) $7/8$
- (D) $1/2$

Q10.

2M [-0.66]

[Engineering Mathematics]

Derivative of x^3 at $x=2$:

- (A) 6
- (B) 12
- (C) 8
- (D) 4

Q11.

2M [-0.66]

[Engineering Mathematics]

Determinant of 2x2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q12.

2M [no -ve]

*[Engineering Mathematics]*Mean of binomial $n=10, p=0.5$:

*Enter your answer here***Q13.**

2M [no -ve]

*[Engineering Mathematics]*Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

*Enter your answer here***Q14.**

1M [-0.33]

[Programming and Data Structures]

Avg hash lookup:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n^2)$

Q15.

1M [-0.33]

[Programming and Data Structures]

Infix to postfix uses:

- (A) Queue
- (B) Stack
- (C) Tree
- (D) Graph

Q16.

1M [-0.33]

[Programming and Data Structures]

Inorder traversal of BST gives:

- (A) Sorted order
- (B) Reverse sorted
- (C) Level order
- (D) Random

Q17.

1M [-0.33]

[Programming and Data Structures]

What is the time complexity of inserting at the beginning of a linked list?

- (A) $O(n)$
- (B) $O(1)$
- (C) $O(\log n)$
- (D) $O(n^2)$

Q18.

1M [-0.33]

[Algorithms]

BFS uses:

- (A) Stack
- (B) Queue
- (C) Heap
- (D) Tree

Q19.

1M [no -ve]

[Algorithms]

Merge sort space:

*Enter your answer here***Q20.**

1M [-0.33]

[Algorithms]

Worst-case time complexity of merge sort:

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $O(\log n)$

Q21.

1M [-0.33]

[Algorithms]

LCS uses:

- (A) Greedy
- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q22.

2M [no -ve]

[Algorithms]

The time complexity of Dijkstra's algorithm using a binary heap is:

*Enter your answer here***Q23.**

1M [no -ve]

[Algorithms]

Kruskal uses:

*Enter your answer here***Q24.**

1M [-0.33]

[Algorithms]

Time complexity of binary search:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(1)$

Q25.

1M [no -ve]

[Operating Systems]

TLB stands for:

Enter your answer here

Q26.

2M [-0.66]

[Operating Systems]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q27.

1M [-0.33]

[Operating Systems]

Round-robin uses:

- (A) Time quantum
- (B) Priority
- (C) Arrival time
- (D) Burst time

Q28.

1M [-0.33]

[Operating Systems]

Which of the following is NOT a necessary condition for deadlock?

- (A) Mutual Exclusion
- (B) Hold and Wait
- (C) Preemption
- (D) Circular Wait

Q29.

1M [-0.33]

[Operating Systems]

Inode contains:

- (A) File name
- (B) Metadata
- (C) File data
- (D) Directories

Q30.

1M [-0.33]

[Operating Systems]

Semaphore with value 1 is:

- (A) Counting
- (B) Binary (mutex)
- (C) Monitor
- (D) Lock variable

Q31.

1M [-0.33]

[Operating Systems]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q32.

1M [no -ve]

[Computer Networks]

OSPF is:

*Enter your answer here***Q33.**

1M [-0.33]

[Computer Networks]

Asymmetric encryption:

- (A) AES
- (B) DES
- (C) RSA
- (D) 3DES

Q34.

1M [no -ve]

[Computer Networks]

Which protocol operates at the application layer of the TCP/IP model?

*Enter your answer here***Q35.**

1M [no -ve]

[Computer Networks]

CSMA/CD is used in:

*Enter your answer here***Q36.**

1M [-0.33]

[Computer Networks]

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q37.

1M [-0.33]

[Computer Networks]

Protocol to find MAC from IP:

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) ICMP

Q38.

1M [no -ve]

[Digital Logic]

Flip-flop stores:

Enter your answer here

Q39.

1M [-0.33]

[Digital Logic]

Full adder has how many inputs:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q40.

1M [-0.33]

*[Digital Logic]*The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q41.

1M [no -ve]

[Theory of Computation]

Which language is regular?

*Enter your answer here***Q42.**

1M [-0.33]

[Theory of Computation]

Halting problem:

- (A) Decidable
- (B) Undecidable
- (C) Semi-decidable
- (D) None

Q43.

1M [-0.33]

[Theory of Computation]

PDA recognizes:

- (A) Regular
- (B) Context-free
- (C) Context-sensitive
- (D) Recursive

Q44.

1M [-0.33]

[Compiler Design]

3-address code operands:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q45.

1M [-0.33]

[Compiler Design]

Lexical analyzer outputs:

- (A) Tokens
- (B) Parse tree
- (C) Intermediate code
- (D) Assembly

Q46.

2M [no -ve]

[CS]

Consider routing table of an organization's router shown below:

Subnet Number	Subnet Mask	Next Hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.0	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
default		R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

*Enter your answer here***Q47.**

2M [-0.66]

[CS]

A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (rounded off to two decimal places) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

- (A) A
- (B) B
- (C) C
- (D) D

Q48.

2M [no -ve]

[CS]

Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____

- (A) A
- (B) B
- (C) C
- (D) D

Q49.

2M [no -ve]

[CS]

Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?

Enter your answer here

Q50.

2M [-0.66]

[CS]

In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square. What is the area (in cm²) of the rectangle PLMN? Note: The figure shown is representative. N P S M Q LR

- (A) 22
- (B) 2
- (C) 8
- (D) 4

Q51.

2M [-0.66]

[CS]

Consider a sequence a of elements $a_0 = 1$, $a_1 = 5$, $a_2 = 7$, $a_3 = 8$, $a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q , both of which are initially empty. I: push the elements of a from a_0 to a_5 in that order into S . II: enqueue the elements of a from a_0 to a_5 in that order into Q . III: pop an element from S . IV: dequeue an element from Q . V: pop an element from S . VI: dequeue an element from Q . VII: dequeue an element from Q and push the same element into S . VIII: Repeat operation VII three times. IX: pop an element from S . X: pop an element from S . The top element of S after executing the above operations is . CS GATE 2023 Computer Science and Information Technology (CS)

- (A) A
- (B) B
- (C) C
- (D) D

Q52.

2M [-0.66]

[CS]

Consider the following C program: `#include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != '\n') fX(); fX(); if(a != '\n') return 0;} putchar(a);}` Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

- (A) The program will not terminate
- (B) The program will terminate with no output
- (C) The program will terminate with 4321 as output
- (D) The program will terminate with 1234 as output

Q53.

2M [-0.66]

[CS]

Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

- (A) A and B are independent.
- (B) A and C are independent.
- (C) B and C are independent.
- (D) $\text{Prob}(B|C) = \text{Prob}(B)$

Q54.

2M [-0.66]

[CS]

Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation on the set A as $a_1 \sim a_2$ if $f(a_1) = f(a_2)$, where $a_1, a_2 \in A$. Let $E = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : E \rightarrow B$ as $F([x]) = f(x)$, for all the equivalence classes $[x]$ in E . Which of the following statements is/are TRUE?

- (A) F is NOT well-defined.
- (B) F is an onto (or surjective) function.
- (C) F is a one-to-one (or injective) function.

(D) F is a bijective function.

Q55.

2M [no -ve]

[CS]

Consider the following definition of a lexical token id for an identifier in a programming language, using extended regular expressions: letter [A-Za-z] digit [0-9] id letter (letter | digit) Which one of the following Non-deterministic Finite-state Automata with ϵ -transitions accepts the set of valid identifiers? (A double-circle denotes a final state) letter letter digit

- (A) A
- (B) B
- (C) C
- (D) D

Q56.

1M [-0.33]

[Digital Logic]

The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q57.

2M [-0.66]

[CS]

The symbol indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

- (A) $(A) (B) (C) \text{ implies } (D)$
- (B) $(A) (B) (C) \text{ implies } (D)$
- (C) $((A) \text{ and } (B)) \text{ implies } (C)$
- (D) $(A \text{ and } B) \text{ implies } C$

Q58.

2M [no -ve]

[CS]

Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i , respectively. Consider the schedule S with four transactions. $S: R_4(x)R_2(x)R_3(x)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)$ Which one of the following serial schedules is conflict equivalent to S ?

Enter your answer here

Q59.

2M [-0.66]

[CS]

Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S . Which ONE of the following is CORRECT?

- (A) Both (i) and (ii) are TRUE
- (B) (i) is TRUE and (ii) is FALSE
- (C) (i) is FALSE and (ii) is TRUE
- (D) Both (i) and (ii) are FALSE

Q60.

2M [no -ve]

[CS]

Consider the following C program: `#include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != '\n') fX(); fX(); if(a != '\n') return 0;} putchar(a);}` Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

Enter your answer here

Q61.

2M [-0.66]

[CS]

The number of distinct minimum-weight spanning trees of the following graph is _____

b 3 c 1 2 2 1 a
d 2 g 2 1 2 1 2 f e 3

- (A) A
- (B) B
- (C) C
- (D) D

Q62.

2M [no -ve]

[CS]

Let $G = (V, E)$ be a context-free grammar in Chomsky Normal Form with $V = \{S, A, B\}$ and containing 10 variable symbols including the start symbol S . The string $w = 303030$ is derivable from S . The number of steps (application of rules) in the derivation is _____

Enter your answer here

Q63.

1M [-0.33]

[General Aptitude]

If $\sin x = -\frac{1}{2}$ for all real values of x , which one of the following statements is true?

- (A) $\sin x = 0$
- (B) $\sin x = 1$
- (C) $\sin x = 1; \cos x = -1$
- (D) $\sin x = 0$

Q64.

2M [no -ve]

[CS]

Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has even weight. Which of the following statements is/are TRUE for every such graph?

- (A) All edges in G have even weight
- (B) All edges in G have even weight OR all edges in G have odd weight
- (C) In each cycle in G , all edges in G have even weight
- (D) In each cycle in G , either all edges in G have even weight OR all edges in G have odd weight

Q65.

2M [no -ve]

[CS]

A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures: P: Unsorted doubly linked list with pointers to the head node and tail node of the list. Q: Min-heap implemented using an array. R: Binary Search Tree. Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size of these data structures?

- (A) P: (1), Q: (1), R: (1)
- (B) P: (1), Q: ($\log n$), R: (1)
- (C) P: (1), Q: ($\log n$), R: (2)
- (D) P: (1), Q: (1), R: ($\log n$)